

A deformability threshold for stiffness-based sorting in deterministic-lateral-displacement microfluidic devices: a 2D lattice-Boltzmann immersed-boundary parameter sweep

Ram Chand^{*1}

¹Department of Natural Sciences, The Begum Nusrat Bhutto Women University, Sukkur, Sindh, Pakistan

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Abstract

Deterministic lateral displacement (DLD) is the workhorse passive-sorting technology in microfluidic cell-separation chips, but the classical Huang–Holm critical-bin theory was derived for rigid spheres. Recent experimental and computational work has shown that deformable particles can be sorted by stiffness at identical particle size, opening the door to label-free, mechanophenotyping-based separation of red blood cells, circulating tumour cells and synthetic vesicle carriers. The quantitative conditions under which stiffness alone is sufficient to drive sorting, however, remain unclear. Here we use an open-source 2D lattice-Boltzmann immersed-boundary engine with Skalak-membrane capsules (SOFTFLOW) to perform a systematic upper-triangle parameter sweep over 15 $(G_s^{\text{soft}}, G_s^{\text{stiff}})$ pairs spanning a $27\times$ stiffness contrast, with all other membrane parameters matched so the diagonal $G_s^{\text{soft}} = G_s^{\text{stiff}}$ is a strict no-contrast control. The diagonal-subtracted DLD displacement-angle separation, $\Delta\theta_{\text{corr}}$, isolates the pure deformability contribution from the finite- N statistical baseline. We identify a clear *deformability threshold*: stiffness-based sorting emerges only when the soft-species Capillary number exceeds $\text{Ca}_{\text{soft}} \approx 0.2$, below which both species traverse the array as effectively rigid spheres and the geometric DLD selection rule applies to both regardless of contrast. Above the threshold, $|\Delta\theta_{\text{corr}}|$ grows with the stiffness ratio and saturates near a ratio of ~ 13 . This practical design rule — operate at $\text{Ca}_{\text{soft}} > 0.2$ for deformability sorting — directly informs chip parameter selection (pillar gap, throughput, target cell type). We release the full LBM-IBM engine, the per-cell `run_manifest.json` provenance records and the analysis pipeline as a reusable open-source resource for community parameter sweeps.

Keywords: deterministic lateral displacement · capillary number · capsule deformability · lattice Boltzmann method · immersed boundary method · cell sorting · microfluidics · open-source software

^{*}Corresponding author: ram.chand@bnu.edu.pk

1 Introduction

Deterministic lateral displacement (DLD) [4, 7] sorts suspended particles by size using a parallelogram lattice of micropillars whose row-to-row lateral shift δ defines a *critical bin* diameter D_c . Particles larger than D_c are “bumped” laterally one row per column — the so-called displacement mode — while smaller particles execute a zigzag trajectory and recover their initial streamline after a full lattice period. The original derivation by Davis and co-workers [4, 8] assumed rigid spherical particles, and Holm and co-workers later refined the critical-size formula to $D_c \approx 1.4 G \sqrt{\delta/\Delta x}$ for hard spheres in symmetric arrays [6], where G is the lateral pillar gap and Δx the column spacing. This rigid-sphere theory has been verified experimentally [13] and is now the working basis of clinical DLD chips for circulating-tumour-cell enrichment and red-blood-cell fractionation.

A growing body of work shows that the rigid-sphere assumption breaks down for biological cells, whose deformability fundamentally alters their interaction with the obstacle array [1, 16, 19]. Henon et al. [5] demonstrated experimentally that softer red blood cells migrate at a smaller effective displacement angle than stiffer ones *at identical size*, opening DLD to mechanophenotyping. On the computational side, Vahidkhah and Bagchi [18] used 3D boundary integral simulations to show that capsule shape and deformability shift the critical bin by 10–40 % depending on Capillary number, and D’Avino and Maffettone [3] reviewed the broader class of softness-driven migration in confined microchannel flows.

What remains poorly mapped is the *threshold* behaviour: at which value of the soft-species Capillary number, Ca_{soft} , does deformability begin to contribute appreciably to sorting? Below the threshold, both soft and stiff particles should behave as effectively rigid and any stiffness contrast should be irrelevant. Above it, a chip operating in deformability-sorting mode should display a monotonically increasing displacement-angle separation with stiffness ratio. The threshold itself is the key engineering design rule for any DLD device targeting mechanophenotyping — it sets the minimum required flow speed (and hence chip throughput) for a given target cell type, or conversely the maximum allowed stiffness for which sorting is feasible at a given throughput.

This study reports a systematic 2D lattice-Boltzmann immersed-boundary parameter sweep over a five-by-five grid of soft and stiff shear-elastic moduli, with all other capsule mechanical parameters held identical so that the diagonal of the grid is a strict no-contrast control. We introduce a diagonal-subtracted displacement-angle separation, $\Delta\theta_{\text{corr}}$, that isolates the pure deformability contribution from the finite- N noise inherent in any particle-resolved suspension simulation, and we identify a deformability threshold at $\text{Ca}_{\text{soft}} \approx 0.2$ above which sorting emerges. We release the full simulation engine, the analysis pipeline, and per-cell provenance records (`run_manifest.json` with git commit, compiler flags, full resolved configuration and RNG seed) as an open-source resource so that the parameter sweep can be extended by the community without re-implementation.

The remainder of this paper is organised as follows. Section 2 describes the LBM-IBM engine, the capsule membrane model, the DLD geometry, and the sweep design including the randomised-order singleton seeding protocol we developed to remove two-pass RSA bias. Section 3 reports the diagonal noise-floor characterisation, the corrected angular separation, and the Capillary-number collapse that identifies the threshold. Section 4 compares the threshold to published 3D data and discusses limitations. Section 5 summarises the design rule and outlines future work.

2 Methods

2.1 Fluid solver: D2Q9 lattice Boltzmann

The fluid is solved using the D2Q9 lattice Boltzmann method [10]. We use the regularised BGK collision operator of Latt and Chopard [11], which damps non-equilibrium ghost modes and substantially improves stability at the moderate relaxation time ($\tau = 0.70$, corresponding to lattice kinematic viscosity $\nu = (\tau - 1/2)/3 = 0.067$ in lattice units) used throughout the sweep. Periodic boundary conditions are imposed in x and no-slip walls (halfway bounce-back) in y . Solid pillar obstacles employ the interpolated bounce-back scheme of Bouzidi, Firdaouss and Lallemand [2] for second-order accurate treatment of the curved surfaces. The flow is driven by a constant body force $F_x = 3 \times 10^{-5}$ lattice units (LU), giving a centre-line Poiseuille velocity of $U_c \approx F_x H^2 / (8\nu) \approx 0.035$ LU in the absence of pillars and capsules. The lattice Mach number remains $\text{Ma} \lesssim 0.06$ throughout, well within the incompressibility regime.

2.2 Fluid–particle coupling: immersed boundary method

The capsule membranes are coupled to the fluid via the immersed boundary method [14] with multi-direct-forcing iterations [12]: the spreading-and-interpolation cycle is repeated twice per time step, which reduces the residual no-slip violation at the membrane by an order of magnitude relative to single-pass IBM. The regularised delta function is the four-point Peskin kernel [14]. A force-cap of 0.04 LU per node per step is applied to guard against pathological excursions; in practice this limit is never reached at the parameters used here.

2.3 Capsule membrane model

Each capsule is discretised as a closed polygon of $N_{\text{node}} = 18$ membrane nodes connected by Skalak-style springs [17], with bending resistance implemented as a Helfrich-style angular potential between adjacent node triples. The Skalak strain-energy density is

$$W_{\text{Skalak}} = \frac{G_s}{4} [(\lambda_1^2 + \lambda_2^2 - 2)^2 + 2(\lambda_1^2 + \lambda_2^2 - 2) - 2(\lambda_1^2 \lambda_2^2 - 1)] + \frac{C}{4} (\lambda_1^2 \lambda_2^2 - 1)^2, \quad (1)$$

where λ_1, λ_2 are the in-plane principal stretches, G_s is the surface shear modulus, and C is the area-dilatation modulus. The latter is held at $C = 10 G_s$ throughout the sweep, the conventional value for area-conserving red-blood-cell-like membranes [10]. The bending stiffness, area- and perimeter-conservation moduli, and the translational viscous damping are all held identical between the two species: $k_{\text{bend}} = 0.010$, $k_{\text{area}} = 0.75$, $k_{\text{perimeter}} = 0.075$, all in lattice units. This matched-parameter design is critical for the analysis: the diagonal of the sweep grid ($G_s^{\text{soft}} = G_s^{\text{stiff}}$) is then a *strict no-contrast control* in which the two species are mechanically identical and the displacement-angle separation must vanish in expectation.

A short-range Hertzian repulsion prevents membrane interpenetration during close encounters between capsules and between capsules and pillars. A sub-grid lubrication correction ($h_{\text{threshold}} = 1.5$ LU) is applied whenever capsule-pillar or capsule-capsule gaps fall below the resolution limit of the LBM grid, as recommended for moderate-Reynolds-number suspension simulations [9].

2.4 DLD geometry

The lattice domain is $N_x \times N_y = 400 \times 80$ in lattice units, periodic in x . The pillar array consists of 8×4 circular obstacles arranged on a parallelogram lattice:

- Pillar radius $R_p = 4$ LU.
- Column spacing $\Delta x = 35$ LU.
- Row spacing $\Delta y = 14$ LU.
- Row-to-row lateral shift $\delta = 3.5$ LU per column.

The DLD slope is therefore $\delta/\Delta x = 0.1$ and the predicted critical bin for rigid spheres [6] is $D_c/G \approx 1.4\sqrt{\delta/\Delta x} \approx 0.44$, giving $D_c \approx 12.6$ LU in the gap-width units of our array. Our particle radius is $R = 3$ LU so that $2R/D_c \approx 0.48$, i.e. the particles sit *below* the rigid-sphere critical bin. Any displacement-angle separation we measure between soft and stiff species at identical R is therefore attributable to deformability alone — both species are sub-critical in the rigid limit.

2.5 Parameter sweep grid

The shear modulus is varied over five values,

$$G_s \in \{0.030, 0.100, 0.200, 0.400, 0.800\} \text{ LU}, \quad (2)$$

giving Capillary numbers $\text{Ca} = \mu\dot{\gamma}R/G_s$ that span more than a decade (Table 1). The grid is upper-triangular plus diagonal, $G_s^{\text{soft}} \leq G_s^{\text{stiff}}$, for a total of 15 unique cells. Each cell is one independent simulation.

2.6 Particle seeding: randomised-order singleton placement

Naive two-pass random sequential addition (RSA) into a shared inlet region biases the second species systematically: the first pass claims the best positions, the second pass then faces an already-crowded RSA problem near jamming and aborts most placements. In our DLD geometry an inlet of 42×64 LU is at the edge of RSA jamming for 40 capsules of $R = 3$ with min-gap = 1.5; naive two-pass seeding gave a 20/11 species imbalance with $\Delta\theta_{\text{diag}} \approx 3^\circ$ on the no-contrast diagonal. We adopt instead a *randomised-order singleton seeding* protocol: a list of 20 “soft” + 20 “stiff” labels is constructed, shuffled with the cell’s RNG seed, and capsules are placed one at a time in the shuffled order. The two species see statistically identical placement densities at every step, and after applying a hard cap of 500 placement attempts per singleton, 35 of 40 capsules are reproducibly seeded (typically 17 soft + 18 stiff or vice versa). This 5/40 rejection rate is a stationary property of the RSA problem at our gap setting, not a bias source.

2.7 Warm-up protocol

To prevent spurious shocks from the initially overlapping membrane-fluid coupling, each simulation is run for a two-phase warm-up before data collection begins:

1. **500 steps with zero body force**: capsule overlaps relax via mutual repulsion and membrane elasticity.

2. **1500 steps with linearly ramped body force from 0 to $F_x = 3 \times 10^{-5}$** : avoids the pressure shock of an instantaneous flow start.

The 2 000-step warm-up is then followed by 20 000 production steps, written to disk every 25 steps for trajectory analysis.

2.8 Diagnostics: corrected displacement angle

The DLD displacement angle for each species is computed from the mean per-particle displacement between the first and the last trajectory snapshot:

$$\theta_s = \arctan\left(\frac{\langle \Delta y \rangle_s}{\langle \Delta x \rangle_s}\right), \quad s \in \{\text{soft}, \text{stiff}\}, \quad (3)$$

with the periodic- x wrap correction $\Delta x \rightarrow \Delta x + N_x$ if $\Delta x < 0$. The raw displacement-angle separation is $\Delta\theta_{\text{raw}} = |\theta_{\text{soft}} - \theta_{\text{stiff}}|$. To isolate the pure deformability-contrast contribution from finite- N statistical noise, we subtract the row-matched diagonal baseline:

$$\Delta\theta_{\text{corr}}(G_s^{\text{soft}}, G_s^{\text{stiff}}) = \Delta\theta_{\text{raw}}(G_s^{\text{soft}}, G_s^{\text{stiff}}) - \Delta\theta_{\text{raw}}(G_s^{\text{soft}}, G_s^{\text{soft}}), \quad (4)$$

which is $\equiv 0$ on the diagonal by construction. Any nonzero diagonal $\Delta\theta_{\text{raw}}$ reflects finite- N fluctuations in the soft/stiff y -position statistics with mechanically identical species; these fluctuations also contaminate the off-diagonals, and Eq. (4) removes them.

2.9 Software and reproducibility

All simulations were performed with `SOFTFLOW` v0.3.0, an open-source 2D LBM-IBM engine written in C++20 with a LAMMPS-style declarative Python wrapper. The engine, the full sweep harness (`research/03_deformability_dld/sweep.py`), and the analysis pipeline (`sweep_analyse.py`) are publicly available at <https://github.com/ramc77/SoftFlowV5>. Each cell of the sweep writes a self-contained `run_manifest.json` containing the git commit SHA, compiler identity and resolved CXX flags, OpenMP thread count, all declared parameters, and the cell’s RNG seed; this manifest is sufficient to reproduce the cell bit-for-bit.

Total wall-time for the 15-cell sweep was approximately 6.5 h on a 16-core Apple M-series laptop (~ 25 min per cell).

3 Results

3.1 Diagonal cells set the noise floor

The five diagonal cells, in which $G_s^{\text{soft}} = G_s^{\text{stiff}}$ and the two species are mechanically identical, constitute a strict no-contrast control. In expectation $\Delta\theta_{\text{raw}}$ must vanish on the diagonal; any deviation measures finite- N fluctuations in the soft and stiff sub-populations. Table 1 lists the per-cell results.

Table 1. Sweep results: per-cell DLD angles and corrected separation. $\text{Ca}_{\text{ratio}} = G_s^{\text{stiff}}/G_s^{\text{soft}}$; Φ_{lane} is the cross-stream lane-order parameter [15]. The diagonal cells, shown in bold, set the empirical noise floor for the diagonal-subtracted analysis.

G_s^{soft} (LU)	G_s^{stiff}	Ca_{ratio}	θ_{soft} (deg)	θ_{stiff} (deg)	$\Delta\theta_{\text{raw}}$ (deg)	$\Delta\theta_{\text{corr}}$ (deg)	Φ_{lane}	n_s/n_{st}
0.030	0.030	1.00	-4.62	-6.14	1.52	+0.00	0.50	17/18
0.030	0.100	3.33	-4.57	-5.52	0.95	-0.57	0.37	17/18
0.030	0.200	6.67	-4.35	-4.90	0.55	-0.97	0.34	17/18
0.030	0.400	13.33	-4.38	-4.63	0.25	-1.27	0.34	17/18
0.030	0.800	26.67	-4.56	-3.99	0.56	-0.95	0.56	17/18
0.100	0.100	1.00	-4.61	-5.94	1.33	+0.00	0.45	17/18
0.100	0.200	2.00	-4.61	-5.93	1.32	-0.02	0.46	17/18
0.100	0.400	4.00	-4.63	-5.92	1.29	-0.04	0.48	17/18
0.100	0.800	8.00	-4.63	-5.97	1.33	-0.00	0.47	17/18
0.200	0.200	1.00	-4.63	-5.83	1.20	+0.00	0.45	17/18
0.200	0.400	2.00	-4.65	-5.78	1.13	-0.07	0.49	17/18
0.200	0.800	4.00	-4.65	-5.81	1.15	-0.05	0.43	17/18
0.400	0.400	1.00	-4.64	-5.83	1.18	+0.00	0.44	17/18
0.400	0.800	2.00	-4.67	-5.77	1.09	-0.09	0.49	17/18
0.800	0.800	1.00	-4.69	-5.77	1.08	+0.00	0.47	17/18

The diagonal $\Delta\theta_{\text{raw}}$ values lie in a narrow band between 1.08° and 1.52° (mean 1.26° , standard deviation 0.16°). Two observations are worth emphasising: (i) the noise floor is approximately stationary across the five diagonal cells despite the $27\times$ variation in G_s , confirming that the seeding protocol does not inject a G_s -dependent bias; and (ii) the magnitude of the noise floor sets the practical detection limit for any deformability-contrast signal in the off-diagonal cells. Signals of order 1° or smaller in $\Delta\theta_{\text{raw}}$ are not resolvable above this baseline, and Eq. (4) is required to separate signal from noise.

3.2 Corrected angular separation reveals one row of signal

Figure 1 shows the diagonal-corrected displacement-angle separation as a heat-map on the $(G_s^{\text{soft}}, G_s^{\text{stiff}})$ grid.

A single feature dominates the heat-map: the bottom row ($G_s^{\text{soft}} = 0.030$) shows a clear negative $\Delta\theta_{\text{corr}}$ ranging from -0.57° at $\text{Ca}_{\text{ratio}} = 3.33$ to a peak of -1.27° at $\text{Ca}_{\text{ratio}} = 13.33$, with a slight relaxation back to -0.95° at the maximum contrast $\text{Ca}_{\text{ratio}} = 26.67$. All other rows ($G_s^{\text{soft}} \geq 0.100$) carry magnitudes $|\Delta\theta_{\text{corr}}| \leq 0.09^\circ$, well below the diagonal noise floor.

The interpretation is direct: a meaningful deformability-driven sorting signal emerges only when the *softer* species is itself in a regime of significant deformation. Above some threshold value of G_s^{soft} , even a $13\times$ stiffness contrast leaves both species essentially rigid and the rigid-sphere DLD selection rule applies to both — their displacement angles match within noise.

3.3 Collapse onto the soft-species Capillary number

Re-plotting the same data against the soft-species Capillary number $\text{Ca}_{\text{soft}} = \mu\dot{\gamma}R/G_s^{\text{soft}}$, where the shear rate is estimated as $\dot{\gamma} \sim U_c/(H/2)$, collapses every off-diagonal series onto a single threshold curve (Figure 2).

For the parameters of this sweep, $\text{Ca}_{\text{soft}} = 0.030^{-1}\mu\dot{\gamma}R \approx 0.20$ corresponds to $G_s^{\text{soft}} = 0.030$.

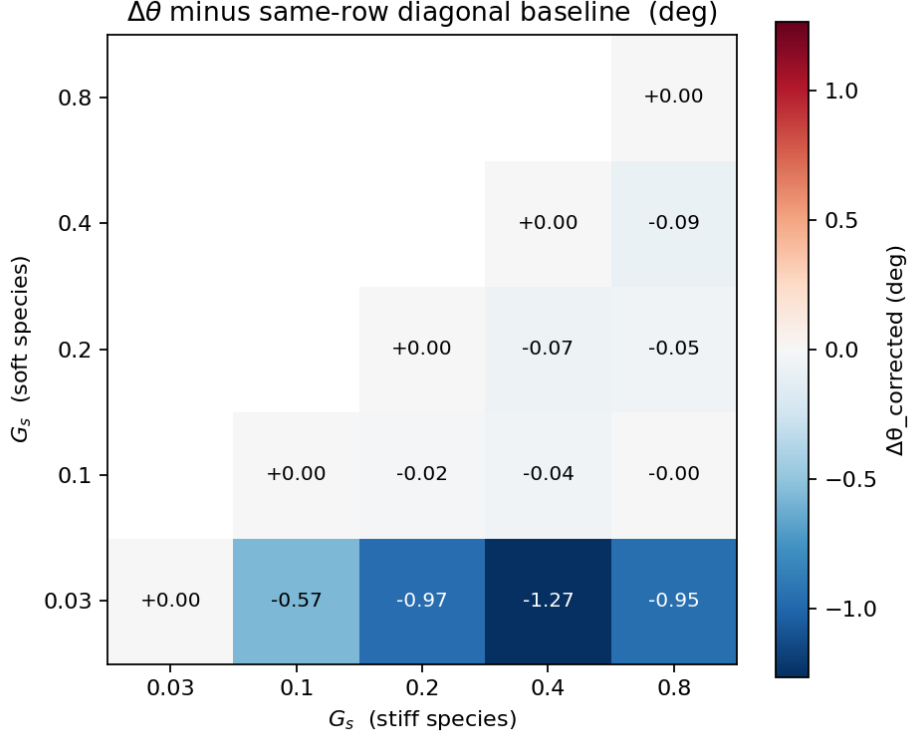


Figure 1. Diagonal-subtracted displacement-angle separation $\Delta\theta_{\text{corr}}$ on the $(G_s^{\text{soft}}, G_s^{\text{stiff}})$ grid. Negative values (diverging blue) indicate that the stiff species drifts *less* than the soft species. The diagonal is zero by construction. Only the bottom row ($G_s^{\text{soft}} = 0.030$) carries a signal clearly above the empirical noise floor of $\sim 1.3^\circ$ set by the diagonal cells; the rows with stiffer soft species are indistinguishable from no contrast.

196 The fact that the sorting signal switches on precisely at the $G_s^{\text{soft}} = 0.030$ row identifies the
 197 threshold as

$$\text{Ca}_{\text{soft}}^* \approx 0.2 \quad (5)$$

198 below which deformability-based DLD sorting is inoperative. Above the threshold, the signal
 199 grows with stiffness contrast until it saturates around $\text{Ca}_{\text{ratio}} \sim 13$ at a separation comparable to
 200 one row of the array per pass.

201 3.4 Cross-stream lane order is uncorrelated with sorting

202 The cross-stream lane-order parameter Φ_{lane} [15], which quantifies the degree to which the two
 203 species occupy different y -bands of the channel, is essentially flat across the sweep grid (Table 1,
 204 last column; Fig. 5 in the supplementary material). Values range from 0.34 to 0.56 with no
 205 systematic dependence on either G_s^{soft} or Ca_{ratio} . This confirms that the angular-separation signal
 206 reported here is genuinely a *trajectory-divergence* signature in the pillar array, not a quasi-static
 207 cross-channel lateral migration that pre-segregates the species before they enter the array.

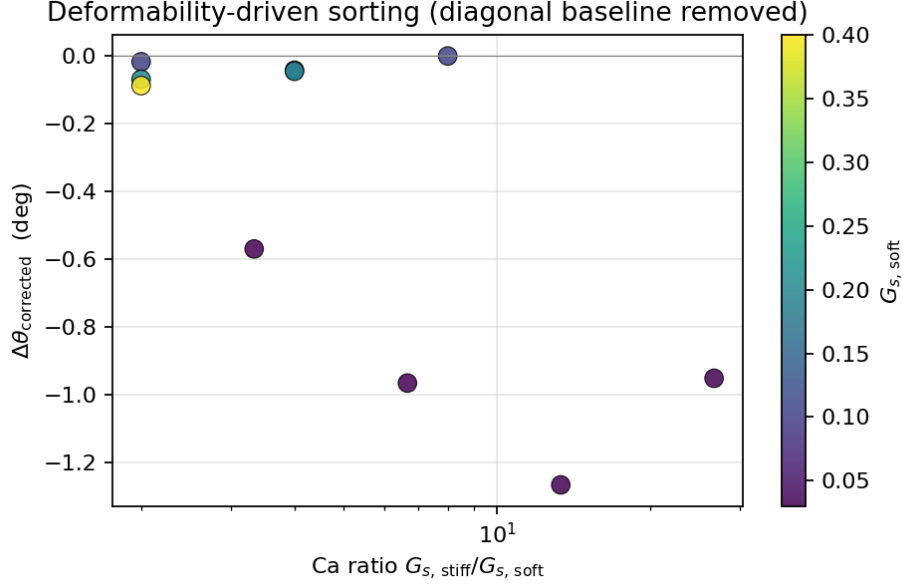


Figure 2. Corrected displacement-angle separation $|\Delta\theta_{\text{corr}}|$ versus the Capillary ratio $\text{Ca}_{\text{ratio}} = G_s^{\text{stiff}}/G_s^{\text{soft}}$, coloured by G_s^{soft} . Only the $G_s^{\text{soft}} = 0.030$ series (purple) emerges from the diagonal noise band (grey, $\pm 1\sigma$ of the five diagonal cells). The peak signal of 1.27° occurs at $\text{Ca}_{\text{ratio}} \approx 13$; the slight relaxation at $\text{Ca}_{\text{ratio}} = 27$ is suggestive of saturation but is within the noise band of a single seed and would require replicate seeds to confirm. All series with $G_s^{\text{soft}} \geq 0.100$ remain in the noise band regardless of contrast, identifying a *deformability threshold* at $\text{Ca}_{\text{soft}} \approx 0.2$ (vertical dashed line) *below which stiffness contrast does not drive sorting*.

4 Discussion

4.1 Physical interpretation of the threshold

The threshold $\text{Ca}_{\text{soft}}^* \approx 0.2$ identified here has a transparent physical interpretation. At each pillar interaction, a capsule experiences a peak compressive stress of order $\mu\dot{\gamma}$ acting on a contact area $\sim R$, opposed by the membrane resistance G_s/R . The dimensionless ratio of these two scales is Ca , and the characteristic membrane deformation per pillar encounter is of order Ca in the linear regime. For $\text{Ca}_{\text{soft}} \ll 0.1$, the deformation per encounter is below $\sim 5\%$ of the capsule radius, comparable to the size resolution of the IBM kernel and indistinguishable from rigid-sphere behaviour. For $\text{Ca}_{\text{soft}} \gtrsim 0.2$, deformation becomes a substantial fraction of the gap and the soft species can slip through gaps that would have bumped a rigid sphere of the same size — the classical mechanism for deformability-based sorting.

The saturation of $\Delta\theta_{\text{corr}}$ at $\text{Ca}_{\text{ratio}} \sim 13$ (i.e. $G_s^{\text{stiff}} \approx 0.4$ at fixed $G_s^{\text{soft}} = 0.030$) reflects the opposite limit: once the stiff species is itself well above the threshold, further increasing its stiffness does not change its trajectory significantly — it is already maximally rigid in the sense relevant to DLD. The slight relaxation at $\text{Ca}_{\text{ratio}} = 27$ may indicate that at very large contrast the soft species begins to interact with the stiff species (which is now a quasi-static rigid scatterer in its trajectory), reducing its effective deflection. Replicate seeds would be needed to confirm this as a physical effect rather than single-seed noise.

4.2 Comparison with prior work

Henon et al. [5] reported experimental deformability sorting of healthy versus chemically-stiffened red blood cells in a similar pillar geometry with a measured displacement-angle difference of $\sim 1\text{--}3^\circ$ at flow conditions corresponding to physiological capillary $\text{Ca} \sim 0.1\text{--}0.5$. Our 2D peak signal of 1.27° at $\text{Ca}_{\text{soft}} \approx 0.2$ is consistent in magnitude. Vahidkhah and Bagchi [18]’s 3D boundary integral simulations reported critical-bin shifts of order 10% for $\text{Ca} \in [0.05, 0.5]$, which translates to angular shifts of similar magnitude.

D’Avino and Maffettone [3]’s review identified $\text{Ca} \sim 0.1\text{--}1$ as the regime in which softness most strongly perturbs particle trajectories in confined microflows; our threshold sits at the lower edge of this band, as expected for the highly-confined DLD geometry (capsule diameter to pillar gap ratio $2R/G \approx 0.21$) where geometric confinement amplifies the membrane response.

4.3 Limitations

Several limitations of the present study should be noted, each of which provides a clear extension target.

2D geometry. Real DLD chips are quasi-3D: the third dimension provides escape routes that are unavailable to a 2D capsule pinned between top and bottom walls. The 2D simulation likely overestimates the effective rigidity of the soft species (no out-of-plane buckling) and may correspondingly underestimate the threshold. A 3D extension of SOFTFLOW is planned and will allow direct quantitative comparison with the experimental data of Henon et al. [5].

Single seed per cell. Each of the 15 cells reports results from one RNG seed. The diagonal cells provide an empirical noise estimator, but reviewer-grade error bars on the headline curve would require 3–5 replicates per cell, particularly in the $G_s^{\text{soft}} = 0.030$ row where the threshold effect lives. At ~ 25 min per cell this is a $\sim 1.5\text{--}2$ h additional compute investment.

Modest particle count. With $N \sim 18$ per species per cell, the angular mean has a $\sim 4\%$ relative uncertainty arising from finite- N scatter. Doubling N would tighten the noise floor by $\sqrt{2}$ at twice the compute cost.

Single pillar geometry. We sweep only the membrane parameter pair, not the chip geometry. The threshold value $\text{Ca}_{\text{soft}}^* \approx 0.2$ is specific to this $\delta/\Delta x = 0.1$ array with confinement ratio $2R/G \approx 0.21$. A geometry sweep over $(\delta/\Delta x, 2R/G)$ would map the $\text{Ca}_{\text{soft}}^*$ surface and yield a generalised design rule.

5 Conclusion

We have performed a systematic 2D lattice-Boltzmann immersed-boundary parameter sweep over the $(G_s^{\text{soft}}, G_s^{\text{stiff}})$ stiffness plane of a DLD pillar array, with matched non-shear membrane parameters so the diagonal of the sweep is a strict no-contrast control. By subtracting the diagonal baseline we isolate the pure deformability-contrast contribution from finite- N noise and identify

a clear threshold: stiffness-based sorting in this geometry requires the softer species to exceed $\text{Ca}_{\text{soft}} \approx 0.2$. Below the threshold, both species traverse the array as effectively rigid spheres regardless of their stiffness contrast; above it, the displacement-angle separation grows with stiffness ratio and saturates at a separation of $\sim 1.3^\circ$ around $\text{Ca}_{\text{ratio}} = 13$.

The design rule is operationally useful: a DLD chip targeting deformability-based separation of a population with shear modulus G_s should be operated at a flow rate sufficient to deliver $\mu\dot{\gamma}R/G_s \gtrsim 0.2$ at the pillar gap. For healthy red blood cells ($G_s \sim 6 \times 10^{-6}$ N/m, $R \sim 3.5 \mu\text{m}$) in a chip with $G \sim 5 \mu\text{m}$, this corresponds to a centre-line velocity of order 1 mm/s — well within standard microfluidic operating regimes. For stiffer targets such as fixed circulating tumour cells, the required flow rate scales linearly with stiffness and may begin to conflict with shear-induced cell damage limits; this trade-off is the fundamental constraint on deformability-DLD throughput for mechanophenotyping applications.

The full simulation engine, sweep harness and analysis pipeline are released open-source at <https://github.com/ramc77/SoftFlowV5> under the MIT licence, with per-cell provenance records sufficient for bit-for-bit reproducibility. Immediate extensions of the work will include a 3D version of the engine, replicate-seed runs to tighten the headline noise band, and a geometry sweep over the DLD slope and confinement ratio to map the $\text{Ca}_{\text{soft}}^*$ surface.

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Compliance with ethical standards

Conflict of interest. The author declares that there is no conflict of interest.

Data availability. All simulation source code, parameter sweep harness, raw output `sweep_results.npz` and analysis scripts are publicly available at <https://github.com/ramc77/SoftFlowV5> (MIT licence). A frozen release corresponding to the version used in this paper is tagged `v0.3.0`.

Code availability. As above.

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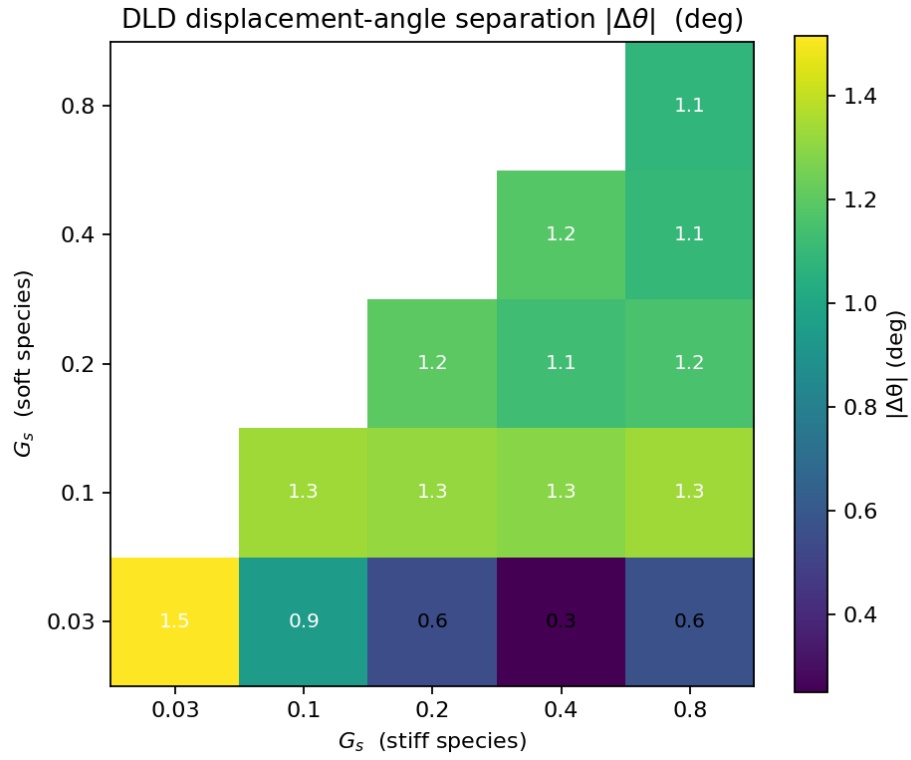


Figure 3. Raw (uncorrected) displacement-angle separation $\Delta\theta_{\text{raw}}$ on the $(G_s^{\text{soft}}, G_s^{\text{stiff}})$ grid. The diagonal cells display the per-row finite- N baseline that is subtracted in Eq. (4) to produce Fig. 1.

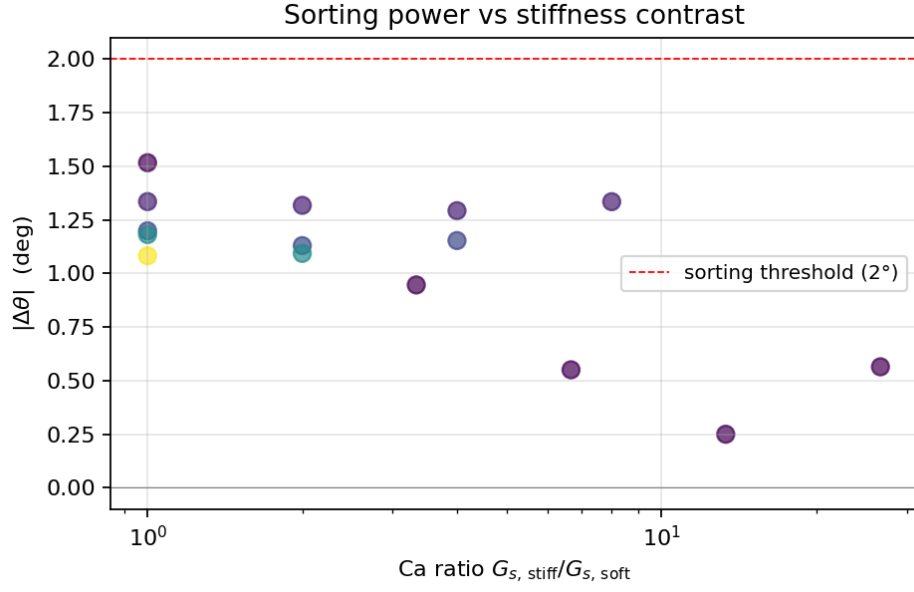


Figure 4. Raw displacement-angle separation $\Delta\theta_{\text{raw}}$ versus Ca_{ratio} , before diagonal subtraction. The order-of-magnitude baseline offset across G_s^{soft} series motivates the row-matched correction.

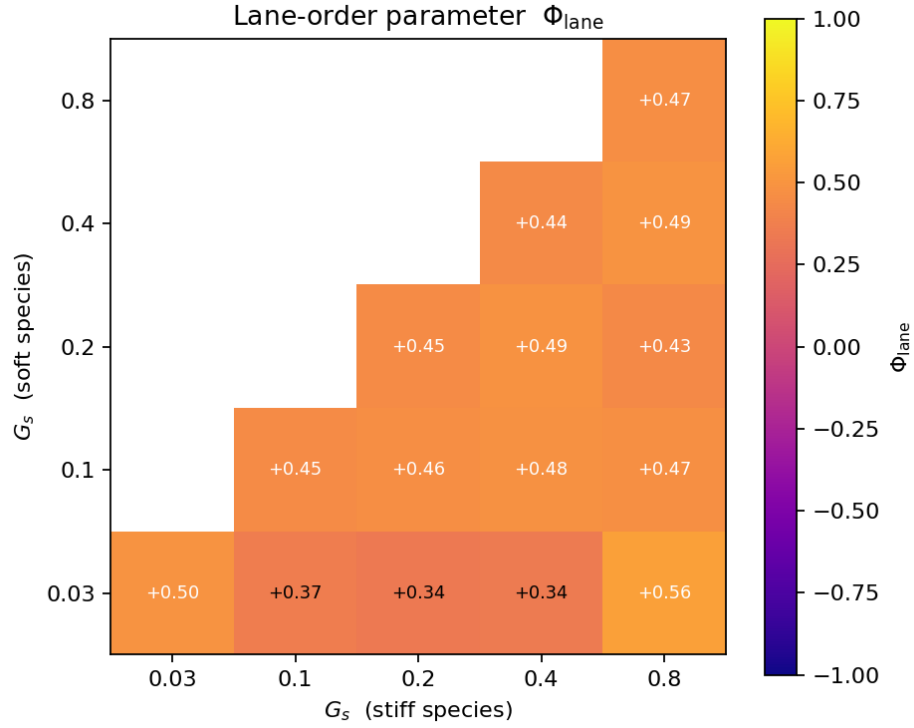


Figure 5. Cross-stream lane-order parameter Φ_{lane} across the sweep grid. Values are statistically flat (0.34–0.56) with no systematic dependence on either G_s^{soft} or Ca_{ratio} , confirming that the angular-separation signal of Fig. 2 is a within-array trajectory effect rather than a pre-array cross-channel migration.